

The uniqueness of differential structures on $\mathbb{R}$ and $\mathbb{R}^2$

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1 Introduction

In this paper, we give $\mathbb{R}$ and $\mathbb{R}^2$ their standard topologies. We define an equivalence relation on atlases of the topological space (X, τ) by diffeomorphisms; we say the equivalence classes of these atlases are differential structures, and in the paper we may mix these two expressions. It is known that for $n \neq 4$, $\mathbb{R}^n$ only has the standard structure, but for $\mathbb{R}^4$, it has uncountably many structures. This paper aims to show the uniqueness statement for $\mathbb{R}$ and $\mathbb{R}^2$. Results for $n \geq 4$ (maybe) will be investigated in future work.

2 Differential structure of $\mathbb{R}^1$

Suffices to show that for every atlas A of $\mathbb{R}$, the manifold $(\mathbb{R}, A)$ is diffeomorphic to $(\mathbb{R}, \text{ST})$, where ST is the standard atlas $\{(\mathbb{R}, \text{id})\}$. Notice that for two atlases A_1, A_2 . $A_1 \cup A_2$ being compatible with A_1 or A_2 implies $(\mathbb{R}, A_1) \cong (\mathbb{R}, A_2)$ via the id map, where $\cong$ means being diffeomorphic, this straight forward from definitions. Our proof can be decomposed into the following steps. Given an atlas A of $\mathbb{R}$

1. Show A is compatible with the chain-like atlas B . By compatibility, we have $(\mathbb{R}, A) \cong (\mathbb{R}, B)$
2. Glue the intervals in B , and show it is compatible with the atlas $C = \{(\mathbb{R}, \sigma)\}$, where $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a homeomorphism. So we get $(\mathbb{R}, B) \cong (\mathbb{R}, C)$
3. Show $(\mathbb{R}, C) \cong (\mathbb{R}, \text{ST})$

1. Chain-like Atlas

Given an atlas $A = \{(U_i, f_i)\}_{i \in S_1}$ of $\mathbb{R}$, since we give $\mathbb{R}$ its standard topology, an open interval J_i is contained in U_i for each i . Restrict the charts to the open intervals, we get a new atlas $\hat{A} = \{(J_i, \hat{f}_i)\}_{i \in I}$ with $\hat{f}_i = f_i|_{J_i}$ compatible with A . Remove some elements in $\hat{A}$ to make it at most countably infinite, we can always achieve this since $\mathbb{R} = \bigcup_{n \in \mathbb{N}} [n, n+1]$ and $[n, n+1]$ is compact, so we can take a countable union of finite subsets in $\hat{A}$.

We define a partial order $\leq_z$ on $S = \{X \in 2^{\hat{A}} \mid \bigcup_{x \in X} x = \mathbb{R}\}$, for $P, Q \in S$ we say $Q \leq_z P \iff Q \subset P$ (the subscript z aims to distinguish this partial order to the usual $\leq$ on $\mathbb{R}$). So Q maybe has less intervals than P but the union of those intervals still covers $\mathbb{R}$. Notice that for a totally ordered subset $S' \subset S$, we always have a lower bound

by taking intersections. That is $\bigcap_{s \in S} s \leq_z s_0$ for all $s_0 \in S'$ (notice that $\bigcap_{s \in S} s$ covers $\mathbb{R}$ since every s does). Using Zorn's lemma by reversing $\leq_z$ to $\geq_z$, we can have a minimal element $m \in S$ such that $m \leq_z s$ for all $s \in S$. Give the corresponding $\hat{f}_k$ s to the intervals in m , making into an atlas m_A .

Definition 1. An atlas B is chain-like if $B = \{(a_i, b_i), \phi_i\}_{i \in S}$, where $S = \mathbb{Z}$ or $\mathbb{N}$ or $-\mathbb{N}$ or $\{-n, -n+1, \dots, m-1, m\}$, and $\forall i, a_i < b_{i-1} < a_{i+1} < b_i$. In other words we have $(a_i, b_i) \cap (a_j, b_j) \cap (a_k, b_k) = \emptyset$ for $i \neq j, j \neq k, k \neq i$.

Lemma 1. m_A is chain-like.

Proof. $m_A = \{(I_k, \hat{f}_k)\}_{k \in K}$ where K is at most countably infinite. To verify the chain-like property, we only need to work with m . For interval $I = (a, b) \in m$, if $J \cap I \neq \emptyset$. Then $J \cap I = (a, c)$ or (d, b) for some $a < c < b$ or $a < d < b$. If not say $J \cap I \subset I$, since J is an interval, we have $J = I \cap J \subset I$, contradicting to minimality since J can be removed from m . Rename I as J_0 , and in the former case when $I \cap J = (a, c)$ for $c > a$, rename J as J_1 , in the latter case rename J as J_{-1} . This renaming procedure is well-defined. Suppose there exists $P \in m$ also with $P \cap I \neq \emptyset$, then $P \cap I = (a, c')$ for some $a < c' < b$ and $P = (c', C')$. Also let $J_1 = (c, C)$. We have $C < C'$ or $C = C'$ or $C > C'$, in all the cases, either J_1 or P can be dropped, contradicting to minimality. Hence there is at most one interval that can be renamed as J_1 , similar for J_{-1} .

Suppose we now have $J_{-n}, \dots, J_p$, if $\bigcup_{i=-n}^p J_i \neq \mathbb{R}$, we need to find some new interval as J_{-n-1} or J_{p+1} . Such interval always exists, towards contradiction, say all other intervals are disjoint from $\{J_{-n}, \dots, J_p\}$. $\bigcup_{i=-n}^p J_i$ is an open interval, since $\bigcup_{i=-n}^p J_i \neq \mathbb{R}$, so $\mathbb{R} \setminus \bigcup_{i=-n}^p J_i$ is not open. But we have $\mathbb{R} = \bigcup_{J \in m} J$, by the disjoint condition, we have

$$\mathbb{R} \setminus \bigcup_{i=-n}^p J_i = \bigcup_{J \in m, J \not\subset \{J_{-n}, \dots, J_p\}} J$$

but the RHS is open, contradiction. If the renaming process terminates, which implies that mp is finite, then done. Otherwise we have to exhaust every element in m to cover $\mathbb{R}$, this is by the minimality of m . So m_A is chain-like. $\square$

We've only performed reordering, renaming, selecting and restricting operations, which makes m_A naturally compatible with A . Let K be an index set for the chain atlas, for $k \in K$, $\hat{f}_k(J_k)$ is also an open interval since $\hat{f}_k$ s are homeomorphisms. We can replace $\hat{f}_k$ by some $g_k = T_k \circ \hat{f}_k$, where T_k is a suitable diffeomorphism (in $(\mathbb{R}, \text{ST})$ sense) that makes g_k orientation preserving and $g_k(I_k) = I_k$. By suitable, we mean an affine transformation between finite length intervals, and a diffeomorphism possibly containing $\ln$ or $\tan$ when infinite length intervals are involved. Notice that the new atlas $m'_A = \{(I_k, g_k)\}_{k \in K}$ is compatible with m_A since the transition maps are T_k s. So far we've shown

$$(\mathbb{R}, A) \cong (\mathbb{R}, m_A) \cong (\mathbb{R}, m'_A)$$

2. Gluing the Chain

Definition 2. For a one dimensional manifold M , and two charts (U_1, f_1) , (U_2, f_2) , we say the charts are joinable if $f_1(U_1) = I_1$, $f_2(U_2) = I_2$, $f_1(U_1 \cap U_2) = f_2(U_1 \cap U_2) = I_1 \cap I_2$ for some open intervals I_1, I_2 such that $I_1 \cap I_2 \neq \emptyset$.

Lemma 2. For two joinable charts (U_1, f_1) , (U_2, f_2) , we can define a new chart (called the joined chart) $(U_1 \cup U_2, f)$ which satisfies the following three properties

- (1). $f(U_1) = f_1(U_1)$, $f(U_2) = f_2(U_2)$
- (2). $(U_1 \cup U_2, f)$ is compatible with the two charts.
- (3). $f|_{U_1 \setminus U_2} = f_1|_{U_1 \setminus U_2}$, $f|_{U_2 \setminus U_1} = f_2|_{U_2 \setminus U_1}$

Proof. Step 1. We wish to construct a f which matches f_1, f_2 on the tails and changes smoothly in $I_1 \cap I_2$. Hence we will first show the following technical construction. Equip (a, b) with the standard atlas, for a diffeomorphism $f : (a, b) \rightarrow (a, b)$ that is orientation preserving ($f' > 0$). There exists a diffeomorphism $g : (a, b) \rightarrow (a, b)$ and some $\delta > 0$ such that

$$g(x) = x \text{ on } (a, a + \delta], \quad g(x) = f(x) \text{ on } [b - \delta, b)$$

It suffices to find some smooth $k : (a, b) \rightarrow [0, \infty)$ such that

$$k((a, b)) \subset (0, \infty), \quad k(x) = 1 \text{ on } (a, a + \delta), \quad k(x) = f'(x) \text{ on } (b - \delta, b), \quad \int_a^b k \, dx = b - a$$

then we can take $g(x) = a + \int_a^x k \, dt$ and restrict x to (a, b) . With the aid of bump functions, we can find δ small and construct two smooth functions $\alpha, \beta : (a, b) \rightarrow [0, \infty)$ such that

$$\alpha = 1 \text{ on } (a, a + \delta], \quad \alpha = 0 \text{ on } [a + 2\delta, b - 2\delta], \quad \alpha = f' \text{ on } [b - \delta, b)$$

$$\beta^{-1}(0) = (a, a + \delta] \cup [b - \delta, b)$$

Now letting

$$k(x) = \alpha(x) + \frac{b - a - \int_a^b \alpha \, dt}{\int_a^b \beta \, dt} \beta(x)$$

The four properties are easy to check.

Step 2. WLOG assume $I_1 = (0, 2), I_2 = (1, 3)$. Since $(U_1, f_1), (U_2, f_2)$ are compatible, we have $g = f_2 \circ f_1^{-1} : (1, 2) \rightarrow (1, 2)$ a diffeomorphism. Assume that it is orientation-preserving. By step 1, we have $F : (1, 2) \rightarrow (1, 2)$ such that $F(x) = x$ on $(1, 1 + \delta]$, $F(x) = g(x)$ on $[2 - \delta, 2)$. we now define f on $U_1 \cup U_2$ by defining it on $U_1 \setminus U_2, U_1 \cap U_2, U_2 \setminus U_1$. We make

$$f|_{U_1 \setminus U_2} = f_1, \quad f|_{U_1 \cap U_2} = F \circ f_1, \quad f|_{U_2 \setminus U_1} = f_2$$

To check the first property of f , by the definition of joinable charts, we have $f(U_1) = f(U_1 \setminus U_2) \cup f(U_1 \cap U_2) = f_1(U_1 \setminus U_2) \cup (F \circ f_1)(U_1 \cap U_2) = (0, 1] \cup (1, 2) = (0, 2) = f_1(U_1)$, and we can check $f(U_2) = (1, 3) = f_2(U_2)$ similarly. The third property is trivial from the definition of f .

Remains to show compatibility between charts, by symmetry, only need to show the compatibility between (U_1, f_1) and $(U_1 \cup U_2, f)$. Since $U_1 \cap (U_1 \cup U_2) = U_1$, we can restrict the second chart to $(U_1, f|_{U_1})$. We have $h = f|_{U_1} \circ f_1^{-1} : (1, 2) \rightarrow (1, 2)$, for $x \in (0, 1 + \delta)$,

we have $h(x) = x$, for $x \in (1, 2)$, we have $f_1^{-1}(x) \in U_1 \cap U_2$, so $h(x) = f|_{U_1}(f_1^{-1}(x)) = (F \circ f_1 \circ f_1^{-1})(x) = F(x)$. The map is well-defined on $(1, 1+\delta)$ since $F|_{(1, 1+\delta)} = \text{id}$. Hence h is a diffeomorphism. $\square$

We have the atlas $m'_A = \{(I_k, g_k)\}_{k \in K}$. By the construction of g_k s, each pair of overlapping charts are joinable, and their transition maps are orientation preserving, hence their joined chart exists. Starting from some single chart, join it with its adjacent charts one at a time following the procedure in the above lemma. The corresponding homeomorphism f keeps changing, index it by $f^1, f^2, f^3, \dots$ (this is not the function composition notation), also index their domains as $V_1, V_2, V_3, \dots$. Note that $V_1 \subset V_2 \subset V_3 \subset \dots \mathbb{R}$. We claim that $\lim_{n \rightarrow \infty} f^n = \mathbf{F}$ is well-defined since for $x \in \mathbb{R}$, V_n s will eventually cover it, and for any V_n , f^n s will eventually be fixed on it. Define a new singleton atlas $C = \{(\mathbb{R}, \mathbf{F})\}$. We have C compatible with m'_A . Pick some chart $(I_k, g_k) \in m'_A$, there exists some V_m such that $I_k \subset V_m$, so the transition map $f \circ g_k$ equals $f^m \circ g_k$, which is a diffeomorphism. So $(\mathbb{R}, m'_A) \cong (\mathbb{R}, C)$

Finally, we show $(\mathbb{R}, C) \cong (\mathbb{R}, \text{ST})$. We have $\mathbf{F} : \mathbb{R} \rightarrow \mathbb{R}$ a homeomorphism. The desired diffeomorphism is just $\mathbf{F}$. Since by definition, $G : X \rightarrow Y$ is a diffeomorphism near $p \in X$ if $\hat{G} = \psi \circ G \circ \phi^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\phi(p) \mapsto (\psi \circ G)(p)$ is diffeomorphism near $\phi(p)$, where ϕ, ψ are maps from the charts in the atlases of X and Y . In this case, we have $X = (\mathbb{R}, \mathbf{F})$, $Y = (\mathbb{R}, \text{ST})$, $n = 1$, $\phi = \mathbf{F}$ and $\psi = \text{id}$. Hence $\hat{G} = \text{id} \circ \mathbf{F} \circ \mathbf{F}^{-1} = \text{id}$, which is obviously a diffeomorphism.

3 Differential structure of $\mathbb{R}^2$

We will show a much stronger theorem.

Theorem 1. *Homeomorphism between two smooth triangulable surfaces can be isotoped into a diffeomorphism.*

By surfaces we mean a two dimensional Hausdorff manifold possibly with boundary. Since $\mathbb{R}^2$ can be triangulated by the vertices set $\{a + bw | a, b \in \mathbb{Z}\}$ with $w = e^{2\pi i/3}$. Given any atlas A on $\mathbb{R}^2$, we can view $(\mathbb{R}^2, A)$ and $(\mathbb{R}^2, \text{ST})$ as two smooth (triangulable) surfaces, since id map is a natural homeomorphism, by the theorem, it can be isotoped into a diffeomorphism, which gives the desired result. We will go through the following steps to show the theorem.

1. Show that we can isotope the homeomorphism into a diffeomorphism near each vertex using the tours trick.
2. Show that the above isotopy can be extended to the neighbourhoods of each edge.
3. Show that the isotopy can be extended to each face.

The above theorems are called handle smoothing theorem in [2]. Note that the above theorem shows that smooth structures of the a topological manifold is not just unique up to diffeomorphism, but they are unique up to isotopy, which is a much stronger result.

3.1 Some Morse Theory

First we need to study some morse theory, and before this, some results in analysis are required.

Theorem 2. (*Partition of unity*) let M be a manifold, and let $O = \{U_i\}_i$ be an open cover of A . Then we have some $F = \{f_i\}_i \subset C^\infty(M, \mathbb{R})$, such that

1. $\forall x \in M, \forall f_i \in F, f(x) \in [0, 1]$
2. $\forall x \in M, \exists V$ open such that $x \in V$ and all but finitely many $f_i \in F$ are 0 on V
3. $\forall x \in M, \sum_i f_i(x) = 1$ (The sum is well-defined by 2)
4. $\forall f_i \in F, \text{supp}(f) = \overline{\{x | f(x) > 0\}} \subset U_i$.

Proof. Standard analysis result. $\square$

Theorem 3. For any manifold M , there exists $f : M \rightarrow [0, \infty)$ such that $f \in C^\infty(M, [0, 1))$ and $f^{-1}([a, b])$ compact. f is called a smooth positive exhaustive function.

Proof. By [3], take a countable selection of precompact open sets U_j from the atlas that covers M . Due to the existence of the partition of unity, define $f(x) = \sum_{j=1}^{\infty} j f_j(x)$, we claim that f is the desired function. First $f(x) \in [0, \infty)$ since $f(x) \geq \sum_j f_j(x) = 1$, it is smooth since locally it's a sum of finitely many f_j s. Notice that we have $\text{supp}(f_j) \subset U_j \subset \overline{U_j}$, pick some $c \in [0, \infty)$ and $N \in \mathbb{N}$ such that $c < N$, and take $x \notin \bigcup_{j=1}^N \overline{U_j}$, we will have

$$f(x) = \sum_{j=N+1}^{\infty} j f_j(x) \geq \sum_{j=N+1}^{\infty} N f_j(x) = N \sum_{j=N+1}^{\infty} f_j(x) \geq N > c$$

so we get $f(x) \leq c \implies x \in \bigcup_{j=1}^N \overline{U_j}$, hence we have $f^{-1}([0, c]) \subset \bigcup_{j=1}^N \overline{U_j}$. RHS is compact and LHS is closed hence LHS is compact. $\square$

Theorem 4. (*Sard's Theorem*) Given manifolds M, N , for $f \in C^\infty(M, N)$, the set $\{f(x) | \text{rank}(df_x) < \dim(N)\}$ (the image of critical values) has zero measure.

Proof. Real analysis result. $\square$

Definition 3. Let M be a n -dimensional manifold, for $f \in C^\infty(M)$, we say f is a Morse function $f(x_1, \dots, x_n) = c + x_1^2 + \dots + x_k^2 - x_{k+1}^2 - \dots - x_n^2$ for some $k \in \{0, \dots, n\}$ in some local coordinates at the all of its critical points. (k is the index at the critical point, it is well-defined since transition maps induce linear isomorphism between tangent spaces but the index of a bilinear form is an invariant under change of basis.)

From elementary vector calculus one can show that under a suitable change of coordinates, any smooth function the critical point p with $\det(Hf_p) \neq 0$, $df_p = 0$ (p is said to be nondegenerate). Can be written in the form of $x_1^2 + \dots + x_k^2 - x_{k+1}^2 - \dots - x_n^2$, hence we will only care about nondegeneracy in the following paragraphs.

Lemma 3. Let M be a compact manifold, for $f : M \rightarrow \mathbb{R}, f \in C^\infty$, suppose f has only nondegenerate critical points, Then for some small enough C^2 approximation of f , say g , it also has no degenerate critical points.

Proof. Notice that an equivalent condition of having only nondegenerate critical points is

$$||df_p|| + ||Hf_p|| > 0, \forall p \in M$$

M is compact, so we have $||df_p + Hf_p|| > m$. If our C^2 approximation ensures $||df_p - Dg_p|| < \epsilon$, $||Hf_p - Hg_p|| < \epsilon, \forall p \in M$ for $2\epsilon < m$, we have $||Dg_p|| + ||Hg_p|| > 0$ everywhere. $\square$

Lemma 4. *For $f : \mathbb{R}^n \rightarrow \mathbb{R}$ smooth, there exists a sequence $\{a_n\} \subset \mathbb{R}^n$ such that for $x \in \mathbb{R}^n$, $f(x) - a_n \cdot x$ has no non degenerate critical points and $a_n \rightarrow 0$*

Proof. we define $G = df$, $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is smooth, by Sard's theorem, the set of regular values of G must be dense in $\mathbb{R}^n$, hence we can pick a sequence $a_n \rightarrow 0$, such that DG having full rank on $G^{-1}(a_n)$. Notice that DG having full rank iff $\det(Hf - DG) \neq 0$, and $x \in G^{-1}(a_n) \iff df_x = a_n \iff d(f - a_n \cdot x) = 0$. Hence we can find a sequence of a_n so that $f - a_n \cdot x$ has no degenerate critical points. $\square$

Theorem 5. *If M is compact, then there always exists a Morse function on M .*

Proof. Say M has atlas $A = \{U_i, f_i\}_i$, we have $M = \bigcup_i U_i$, decompose U_i s into open balls B_j , that is $U_i = \bigcup_j B_j$, we can further assume $\overline{B_j} \subset U_i$ by radius shrinking. We have $M = \bigcup_j B_j$, by compactness and indices reordering, we have $M = \bigcup_{j=1}^N B_j = \bigcup_{j=1}^N U_j$. we have $\overline{B_j} \subset U_j$ and $\overline{B_j}$ being compact, by partition of unity, there exists smooth $h_j : U_j \rightarrow \mathbb{R}$ such that $\text{supp}(h_j) \subset U_j$ and $h_j|_{\overline{B_j}} = 1$ (h_j is called a bump or step function).

We will construct the morse function inductively. Suppose f_{n-1} has no nondegenerate critical points in $\bigcup_{j=1}^{n-1} \overline{B_j}$. Then in local coordinates, let $f_n(x) = f_{n-1}(x) - (a_n \cdot x)h_j(x)$ on U_j , and $f_n = f_{n-1}$ on $M \setminus \text{supp}(h_n)$. Define f_0 to be any smooth function on M . By letting $(a_n \cdot x)h_n(x)$ to be small in C^2 sense, f_n won't create any degenerate critical points in $(\bigcup_{j=1}^{n-1} \overline{B_j}) \cup \overline{B_n} = \bigcup_{j=1}^n \overline{B_j}$. Since $M = \bigcup_{j=1}^N \overline{B_j}$, f_N will be desired function (note that it is smooth since $f_N - f_0$ is a finite sum of smooth perturbations). We call this technique the localised linear tilt. $\square$

Theorem 6. (*Collar neighbourhood theorem*) *For a compact manifold M with boundary ∂M , we have an embedding $i : \partial M \times [0, 1) \rightarrow M$. (Notice that we have the standard structure on $\partial M \times [0, 1)$)*

Proof. $M = (M \setminus \partial M) \cup \partial M$. $M \setminus \partial M$ is open since all the local neighbourhoods are open. Hence ∂M must be closed in M . Since M is compact, ∂M will be compact.

Pick some $x \in \partial M$, and a chart containing it, say (U_x, f_x) , by definition we have $f_x(U_x) \subset \{(x_1, \dots, x_{n-1}, t) \mid t \geq 0\}$ open and $\pi_n(f_x(x)) = 0$. Define a vector field v_x on U_x by $\frac{\partial}{\partial t}$. By the existence of a partition of unity $\{f_x\}_x$, we can extend the vector field to a smooth global vector field on M , by letting $v = \sum_x f_x v_x$. By standard results in ODE theory, for $x \in \partial M$, there exists $t \mapsto \phi_x(t)$ for $t \in [0, \delta(x))$ such that $\phi_x(t)$ is an integral curve ($\phi'_x(t) = v$, $\phi_x(t)$ can be interpreted as flowing the point x by time t along the vector field), and $\phi_x(t)$ depends smoothly on x, t , and $\delta(x)$ depends smoothly on x . By compactness, there exists $\delta = \min_{x \in \partial M} (\delta(x)) > 0$, and we restrict the domain of every integral curve to $[0, \delta)$. Consider the map $\phi : \partial M \times [0, \delta) \rightarrow M$

$$\phi(x, t) = \phi_x(t).$$

ϕ is obviously smooth, and $\phi(\partial M \times \{0\}) = \partial M$. We will shrink δ further to make sure it is a diffeomorphism.

By definition we have $d\phi_{(x,t)} : T_{(x,t)}(\partial M \times [0, \delta]) \rightarrow T_{\phi(x,t)}M$

$$d\phi_{(x,t)}(a, b) = d_x\phi_x(t)(a) + d_t\phi_x(t)(b)$$

We will show $d\phi_{(x,0)}$ has full rank for all $x \in \partial M$. Notice that $d_x\phi_x(0) = \text{id}_x$ and by the definition of an integral curve we have

$$d_t\phi_x(0) \left(\frac{\partial}{\partial t} \right) = v = \sum_y f_y v_y = \sum_y f_y \frac{\partial}{\partial t_y} = \sum_{j=1}^n f_j \frac{\partial}{\partial t_j}$$

fix one neighbourhood U around x , WLOG let the t component of the local coordinate be t_1 . we can express $\frac{\partial}{\partial t_i}$ in terms of $\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_{n-1}}, \frac{\partial}{\partial t_1}$. Since transition maps are diffeomorphisms and boundary charts are defined on $\mathbf{H}^n$, we have $\frac{\partial}{\partial t_i} = \sum_{j=1}^{n-1} \mu_j \frac{\partial}{\partial x_j} + a_i(x) \frac{\partial}{\partial t_1}$ for some positive a_i . So we have $\sum_{j=1}^n f_j \frac{\partial}{\partial t_j} = \sum_j \lambda_j \frac{\partial}{\partial x_j} + (\sum_j f_j a_j) \frac{\partial}{\partial t_1}$, since $\sum_j f_j a_j$ is positive, we have

$$d_t\phi_x(0)(T_0[0, \delta]) = d_t\phi_x(0) \left(\left\langle \frac{\partial}{\partial t} \right\rangle \right) = V + \left\langle \frac{\partial}{\partial t_1} \right\rangle$$

for some $V \leq \left\langle \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_{n-1}} \right\rangle = T_x \partial M$. Notice that $T_{(x,0)} \partial M \times [0, \delta] = T_x \partial M \oplus T_0[0, \delta]$. Hence we have

$$d\phi_{(x,0)}(T_{(x,0)} \partial M \times [0, \delta]) = d_x\phi_x(0)(T_x \partial M) + d_t\phi_x(0)(T_0[0, \delta]) = T_x \partial M + V + \left\langle \frac{\partial}{\partial t_1} \right\rangle = T_{(x,0)} \partial M \times [0, \delta]$$

Hence $d\phi$ has full rank on $\partial M \times \{0\}$. By its continuity and the compactness of ∂M , we have $d\phi$ having full rank on $\partial M \times [0, \delta')$ for some $0 < \delta' < \delta$. Hence ϕ^{-1} is smooth and

$$\partial M \times [0, \delta') \cong \phi(\partial M \times [0, \delta')) \subset M.$$

□

Corollary 1. *For a compact manifold M with boundary ∂M , the morse function can be chosen so that it takes constant value on the boundary.*

Proof. By the collar neighbourhood theorem, we have some $\phi : \partial M \times [0, d) \rightarrow M$ as an embedding such that $\phi(\partial M \times \{0\}) = \partial M$. This makes the collar map $k : \phi(x, t) \mapsto c + t$ smooth. Note that $\phi(\partial M \times [0, d/2])$ is compact, and $\phi(\partial M \times [0, d))$ is open, by the existence of bump functions, we have some $h : M \rightarrow \mathbb{R}$ such that $\text{supp}(h) \subset \phi(\partial M \times [0, d))$ and $h = 1$ on $\phi(\partial M \times [0, d/2])$. Hence $h \cdot k$ defines a global smooth function g on M and agrees with ϕ near the boundary. Notice that Dg on $\partial M \times \{0\}$ is just the projection map π_n , which is not the zero map. By compactness of ∂M , there exists some $\delta > 0$ such that $Dg \neq 0$ on $\phi(\partial M \times [0, \delta))$. $\phi(\partial M \times [0, \delta))$ is open in M , which makes $M' = M \setminus \phi(\partial M \times [0, \delta))$ compact, we can apply the same localised linear tilt technique while keeping the support of each bump function within M' . This makes g morse on M' hence on M since $Dg \neq 0$ on $M \setminus M' = \phi(\partial M \times [0, \delta))$. Notice that we still have $g|_{\partial M} = c$.

Notice that we can strengthen the statement to g taking different constant values on each connected components of ∂M by defining k in the i -th component as $\phi(x, t) \mapsto c_i + t$ for some $c_i \in \mathbb{R}$. □

Theorem 7. *Morse function exists on all non-compact manifolds with boundary.*

Proof. Choose some smooth positive exhaustive $f : M \rightarrow [0, \infty)$. By Sard's theorem, we can pick some $a_n \uparrow \infty$ such that a_n s are regular values of f . Notice that we have $M = \bigcup_i M_i$ where $M_i = f^{-1}([0, a_i])$. We claim that $\partial M_i = f^{-1}(a_i)$. First $\partial M_i \subset f^{-1}(a_i)$ since $f^{-1}([0, a_i])$ is open in M_i . Since a_n s are regular, by implicit function theorem we have $f^{-1}(a_n)$ being a manifold of dimension $n - 1$ (if we assume $\dim(M) = n$). Since M has no boundary, we then have $\partial M_i = f^{-1}(a_i)$.

Consider M_i and M_{i+1} , we can extend the morse function f_i on M_i to some morse function f_{i+1} on M_{i+1} with $f_{i+1}|_{M_i} = f_i$ by induction. First f_1 exists on $f^{-1}([0, a_1])$ since it is compact, and we can make it constant on its boundary $f^{-1}(a_1)$. Assume we have f_i on M_i with $f_i|_{\partial M_i = f^{-1}(a_i)} = c_i$ for some constant c_i . Notice that $f^{-1}([a_i, a_{i+1}])$ is another manifold with boundary being $f^{-1}(a_i) \cup f^{-1}(a_{i+1})$, use the similar construction in the corollary, we have a morse g_i on it with $g_i|_{f^{-1}(a_i)} = c_i$ and $g_i|_{f^{-1}(a_{i+1})} = c_{i+1}$, let $f_{i+1}|_{f^{-1}([0, a_i])} = f_i$ and $f_{i+1}|_{f^{-1}([a_i, a_{i+1}])} = g_i$ gives the desired morse function on $f^{-1}([0, a_{i+1}])$ since f_i matches smoothly with g_i near $f^{-1}(a_i)$ (thanks to the collar neighbourhood theorem). Finally $\lim_{n \rightarrow \infty} f_n = \hat{f}$ gives the desired morse function on M . $\square$

Remark 1. We have $\hat{f}^{-1}([a, b])$ compact for $a < b$ (We call this property properness). We perturb f to form the morse function $\hat{f} = f + p$, and p can be uniformly small, that is $\|p\| < \delta$ for some $\delta > 0$. Hence $f(x) \in [\hat{f}(x) - \delta, \hat{f}(x) + \delta]$, if $\hat{f}(x) \in [a, b]$, then $f(x) \in [a - \delta, b + \delta]$, we have $\hat{f}^{-1}([a, b]) \subset f^{-1}([a - \delta, b + \delta])$. RHS is compact and LHS is closed, hence LHS is compact.

Remark 2. We can make the critical points of the morse function f into different levels. Notice that the critical points are all separated, since locally $f = c + x_1^2 + \dots + x_k^2 - x_{k+1}^2 - \dots - x_n^2$, and the solution to $f = c$ will be locally unique. By properness of f , on each compact set $K \subset M$, we only have finitely many critical points. Since $M = \bigcup_j f^{-1}((-\infty, j])$, we only have countably infinite many critical points. Near the critical point p , pick U that isolates it from other critical points, and we can perturb f to f' by adding $g = \epsilon b$ for some small $\epsilon \in \mathbb{R}$ and for some bump function b such that $b = 1$ on $\bar{B} \subset U$ for some open ball B . Notice that $db \neq 0$ only on some set $V \subset U \setminus B$, $V \setminus B$ is compact, hence $\|df\|$ takes a minimum, so we can pick ϵ small that lifts $f'(p)$ to a distinct level (always exists by Sard's theorem) and also ensures $f' = f + \epsilon b$ won't generate any new critical points on U . We make such perturbations one by one near each critical point, and the desired f' will be $f' = f + \sum_p \epsilon_p b_p$.

Remark 3. Since we obtain the morse function $\hat{f}$ from positive f , and they differ by a small perturbation, we will have $\hat{f}(M) \subset [-\delta, \infty)$ for some δ . By translation $\hat{f}$ can be assumed to be positive.

We now will show the two important theorems in morse theory.

Theorem 8. (Flow theorem) *For manifold M and a proper morse function f , suppose $f^{-1}([0, 1])$ contains no critical points. Then $f^{-1}([0, 1]) \cong C_0 \times [0, 1]$, in particular we will have $f^{-1}(0) \cong f^{-1}(1)$.*

Proof. Denote $f^{-1}([0, 1])$ as A and $f^{-1}(0) = C_0$, $f^{-1}(1) = C_1$, $f^{-1}(l) = C_l$. We will construct a vector field on A and flow everything along it. Suppose A is covered by

$\{U_i\}_{i \in I}$ for some finite set I . U_i is homeomorphic to some open set in $\mathbb{R}^n$, we can induce the standard inner product in $\mathbb{R}^n$ to a smooth inner product g_i on U_i . By partition of unity, $g = \sum_i f_i g_i$ defines a global smooth inner product on A because any convex combination of nondegenerate and positive definite matrices is still nondegenerate and positive definite. By definition we have $df_x \in T_x A^*$. By Riesz representation theorem, we have $df_x = g_x(v_x, \cdot)$ for some $v_x \in T_x A$, where v_x is a smooth global vector field on A since g is smooth. Since f has no critical points in A , the normalised vector field

$$V(x) = \frac{v_x}{g_x(v_x, v_x)}$$

is well-defined and smooth. We have $df_x(V(x)) = g_x(v_x, V(x)) = \frac{g_x(v_x, v_x)}{g_x(v_x, v_x)} = 1$. For each $x \in C_0$, we have some $\phi_x(t)$ as integral curve such that $\phi_x(0) = x$. Consider the map $\phi : C_1 \times [0, 1] \rightarrow A$ such that $\phi(x, t) = \phi_x(t)$. First ϕ is injective by the uniqueness of solutions and ϕ is smooth. Also notice that $\frac{d}{dt} f(\phi(x, t)) = df_{\phi(x, t)}(V(\phi(x, t))) = 1$ and $f(\phi(x, 0)) = 0$ since $\phi(x, 0) = x \in C_0$. Hence for fixed point x we can eventually reach the set C_1 at $t = 1$, by this, we can show the surjectivity of ϕ . Starting at some $p \in A$, flow it backwards to get $p' \in C_0$ with time duration τ , then $p = \phi(p', \tau)$. We will first show $C_0 \cong C_l$ for some small enough l . Similar to the proof of the collar neighbourhood theorem, remains to show $d\phi$ having full rank $C_0 \times [0, l]$. So we only need to look at $C_0 \times \{0\}$. Pick $(x, 0) \in C_0 \times \{0\}$, we still have $d\phi_{(x, 0)}$ being the identity map on the first $n - 1$ coordinates $x_1, \dots, x_{n-1}$, and notice that

$$df_{\phi(x, 0)} \left(\frac{\partial}{\partial x_i} \right) = 0, \quad \forall i \in \{1, \dots, n - 1\}$$

since $T_x C_0 = \left\langle \frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_{n-1}} \right\rangle$.

But we have $d\phi_{(x, 0)} \left(\frac{\partial}{\partial t} \right) = V(\phi(x, 0))$, so $df_{\phi(x, 0)} \left(d\phi_{(x, 0)} \left(\frac{\partial}{\partial t} \right) \right) = 1$, which implies $d\phi_{(x, 0)} \left(\frac{\partial}{\partial t} \right) \notin T_x C_0$, hence $d\phi$ has full rank on $C_0 \times \{0\}$. Again by continuity of $d\phi$ and compactness of C_0 , we have $\phi : C_0 \times [0, l] \rightarrow f^{-1}([0, l])$ being a diffeomorphism for some small l . Since we have $\phi(C_0 \times \{0\}) = C_0$, $\phi(C_0 \times \{l\}) = C_l$ and a diffeomorphism restricting the domain to a submanifold is still a diffeomorphism. We have

$$C_0 \cong C_0 \times \{0\}, \quad C_l \cong C_0 \times \{l\}$$

since $C_0 \times \{0\} \cong C_0 \times \{l\}$ by projection, we have $C_0 \cong C_l$. But we can identify any level set as the zero level and using the same argument as above, we can flow it to a higher level set. Since $[0, 1]$ is compact, we can flow C_0 bits by bits in finitely many times to reach C_1 . In other words we will have a chain of diffeomorphisms

$$C_0 \cong C_{l_1} \cong C_{l_2} \cong \dots \cong C_{l_n} \cong C_1$$

for $l_{i+1} - l_i$ sufficiently small. □

Definition 4. A k -handle is defined as $D^k \times D^{n-k}$ and is denoted as H^k . Furthermore we define $\partial_1 H^k = S^{k-1} \times D^{n-k}$ and $\partial_2 H^k = D^k \times S^{n-k-1}$. Notice that $\partial H = \partial_1 H \cup \partial_2 H$.

Theorem 9. (*Handle Attachment Theorem*) For manifold M and a proper morse function f , suppose $f^{-1}((0, 1))$ contains exactly one index k critical point. Then $f^{-1}((-\infty, 0]) \cup_i H^k \cong f^{-1}((\infty, 1])$ for some smooth embedding $i : \partial_1 H \rightarrow f^{-1}(0)$.

It can be shown that one can smoothen the corners of the attachment, but we omit the details.

Proof. Suppose we have $f(p) = 1/2$ with p being the critical point, by the flow theorem, suffices to show that $f^{-1}((-\infty, 1/2 + \epsilon]) \cong f^{-1}((-\infty, 1/2 - \epsilon]) \cup_i H^k$. Denote $f^{-1}((-\infty, 1/2 + \epsilon])$ as M_1 and denote $f^{-1}((-\infty, 1/2 - \epsilon])$ as M_2 . Pick some chart around p , we can wlog assume $f(x, y) = 1/2 - |x|^2 + |y|^2$, where $(x, y) \in \mathbb{R}^n, x \in \mathbb{R}^k, y \in \mathbb{R}^{n-k}$. Since we can view H^k as the thickened version of D^k , we will exhibit a smooth retraction to $M_1 \cup_i H^k$ by flowing along a vector field.

Define Core as $\{(x, 0) : f(x, 0) = r - |x|^2 \geq r - \epsilon\} = \{(x, 0) : |x|^2 \leq \epsilon\} \cong D^k \times \{0\}$. Define two regions $R_1 = \{(x, y) : |x|^2 + |y|^2 < 2\epsilon\}$ and $R_2 = \{(x, y) : |x|^2 + |y|^2 < \delta\}$ for $2\epsilon < \delta$. We will construct a vector field which is $(2x, -2y)$ outside of R_2 , and $(0, -2y)$ inside of R_1 . Define

$$v = \begin{cases} (2\beta(x, y)x, -2y) & \text{on } R_2 \\ (2x, -2y) & \text{outside } R_2 \end{cases}$$

with $\beta : \mathbb{R}^m \rightarrow [0, 1]$ equal to 0 on R_1 and 1 outside R_2 . Let

$$\varphi(t, z) = \text{flow of } \frac{v}{|v|} \text{ for time } t \text{ starting from } z \in M_2,$$

Define $T(z) =$ time required by $\frac{v}{|v|}$ to reach $M_1 \cup \text{Core}$. Notice that if $z \in M_1$ then $T(z) = 0$, $\varphi(0, z) = z$. Moreover, $T(z)$ is finite for $z \in R_2^c \cap \{|x| > \epsilon\}$ since $d\left(\frac{v}{|v|}\right) = \frac{-4|y|^2 - 4|x|^2}{\sqrt{4|y|^2 + 4|x|^2}} = -2\sqrt{|x|^2 + |y|^2}$, and $z \in R_2^c$ implies $\frac{d}{dt}|x| > 0 \implies |x| > \epsilon$ and before reaching M_2 , we have $-|x|^2 + |y|^2 > -\epsilon^2 \iff |y|^2 > |x|^2 - \epsilon^2$, hence we have a lower bound for $|y|$, hence we have a uniform bound $-\sqrt{|x|^2 + |y|^2} < -c$ for some $c > 0$. Hence after sufficiently large time t , we have the region $R_2^c \cap \{|x| > \epsilon\}$ retracts to M_1 . Inside R_2 , since $\beta \rightarrow 0$, the time to reach $M_1 \cup \text{Core}$ might be ∞ , since we expect an exponential decay of $|y|$, but for large time t , $M_2 \cap R_2$ retracts to a thickened neighbourhood of **Core**, which is H^k smoothly attached to M_1 . Hence, we have $z \mapsto \varphi(t, z)$ for t large smoothly deform retracts M_2 onto $M_1 \cup H$. □

Definition 5. Given a manifold M and a positive proper morse function f with critical points all on different level sets. Pick some $\{a_n\}$ with $a_n \uparrow \infty$ and regular. Identify each connected components of $f^{-1}(a_n)$ as vertical edges in $\mathbb{R}^2$. For edges e_1, e_2 obtained from $f^{-1}(a_n)$ and $f^{-1}(a_{n+1})$, connect the lower vertex of e_2 to the upper vertex of e_1 if $f^{-1}(a_n)$ and $f^{-1}(a_{n+1})$ are connected in $f^{-1}([a_n, a_{n+1}])$. The resultant graph is called the Reeb graph of M (Notice that the connected components of $f^{-1}([a_n, a_{n+1}])$ are now identified as vertices).

By the above definition we will have a continuous surjective quotient map $q : M \rightarrow G$.

Lemma 5. *For a two dimensional orientable manifold M and its Reeb graph G . Suppose for each edge $e \in G$, we have $q^{-1}(e)$ diffeomorphic to (S^1, ST) . Then for any vertex $v \in G$ with $q^{-1}(v) \subset M$. We have $\deg(v) = 1, 2, 3$ iff $q^{-1}(v)$ contains a maximum/minimum point, no critical points, or a saddle point, respectively.*

Proof. WLOG by scaling, let $q^{-1}(v) \subset f^{-1}([0, 1])$ for a proper morse function f with critical points on distinct levels. WLOG let the critical point (if there is any) be on the level $1/2$. Let F be the restriction of f on this component, which is still morse, then we have $q^{-1}(v) = F^{-1}([0, 1])$. It suffices to only show necessity since $\text{card}\{1, 2, 3\} = 3 = \text{card}\{\text{max/min, no critical points, saddle}\}$

Suppose $q^{-1}(v)$ contains a minimum point $p \in f^{-1}(1/2)$. Then we have $q^{-1}(v) \cong (F^{-1}(0) \times [1/2 + \epsilon, 1]) \cup_i (D^2 \times \{0\})$ with $i : S^1 \times \{0\} \rightarrow F^{-1}(1/2 + \epsilon)$ for some ϵ small. The center of the disk D^2 is p and all the points in the complement will be in a level higher than $1/2$, if not, by intermediate value theorem, we will create a minimum, violating the assumption of $q^{-1}(v)$ containing at most one critical point. Hence $(F^{-1}(0) \times [1/2 + \epsilon, 1]) \cup_i (D^2 \times \{0\})$ is disconnected from the set $F^{-1}(0)$, this implies $\deg(v) = 1$. By symmetry, this is also true when p is a maximum point.

When $q^{-1}(v)$ contains no critical point, then by the flow theorem we can flow from $F^{-1}(0)$ to $F^{-1}(1)$, hence v has two edges attached, that is $\deg(v) = 2$.

If $q^{-1}(v)$ contains a saddle point then by the attachment theorem we have $q^{-1}(v) = F^{-1}([0, 1]) \cong F^{-1}([0, 1/2 - \epsilon]) \cup_i D^1 \times D^1$ for some $i : S^0 \times D^1 \rightarrow F^{-1}(1/2 - \epsilon)$. We can two possible attachments of the 1-handle $D^1 \times D^1$. Identify S^0 as $\{0, 1\}$, the map i restricted to $\{0\} \times D^1$ and $\{1\} \times D^1$ either have the same or the opposite orientation. The two cases correspond to attaching a regular strip or a Mobius band. We shall discard the latter since M is assumed to be orientable. After attaching the 1-handle, $q^{-1}(v)$ will be a pair of pants. Note that if the handle is attached to two different components of $F^{-1}(1/2 - \epsilon)$, the pants will have the normal orientation, that is v having two edges connecting to it down below and one in the above. If the handle is attached to only one component of $F^{-1}(1/2 - \epsilon)$, then we will have the pants and the edges rotated by π , that is two edges on the above and one down below. In either case we get $\deg(v) = 3$.

We call v as a leaf vertex, transition vertex and branch vertex iff $\deg(v) = 1, 2, 3$, respectively. $\square$

Lemma 6. *Let M be a two dimensional manifold and G be its Reeb graph, assume all edges corresponds to a submanifold in M that is diffeomorphic to $(S^1 \times [0, 1], ST)$. Suppose we have a subgraph $G_0 \subset G$ consists of consecutive vertices $\{v_1, \dots, v_n\}$ (and edges connecting them), with v_1 being a branch vertex, v_n being a leaf vertex, and $v_2, \dots, v_{n-1}$ all transition vertices. Then we have $\cup_{i=1}^n q^{-1}(v_i) \cong (S^1 \times [0, 1], ST)$ (We call G_0 a branching stick).*

Proof. By the previous lemma we have the branch edge containing one saddle point and the leaf edge containing a maximum or minimum point. By the attachment theorem we have

$$q^{-1}(v_1) \cong (S^1 \times [0, 1]) \cup_f H^1$$

The attachment is smooth and splits the boundary at the top into two S^1 s, since $H^1 = D^1 \times D^1$ creates two copies of D^1 to the boundary. When growing the tree from v_2 to v_{n-1} , by the flow theorem, we are attaching $S^1 \times [0, n-2]$ to one of the circles. Again by the attaching theorem, when we reach v_n , we glue a standard disk $H^2 \cong D^2$ to the

top boundary of $S^1 \times [0, n-2]$. Notice that by viewing $S^1 \times [0, n-2] \cup_f D^2$ in three dimensional space, by perturbing it into a cone with the round vertex and projecting it to $\mathbb{R}^2$, we see it's diffeomorphic to the standard D^2 . So we have

$$\bigcup_{i=1}^n q^{-1}(v_i) \cong (S^1 \times [0, 1] \cup_{i_1} H^1) \cup_{i_2} D^2$$

note that D^2 is glued along some $S^1 \cong D^1 \cup_{S^0} D^1$ where one of the $D^1 \subset \partial_2 H^1$. So $H^1 \cup_{i_2} D^2$ is diffeomorphic to some D^2 gluing to $D^1 \times D^1$ along the boundary piece that is diffeomorphic to D^1 , the glueing makes it into another standard D^2 , denote this disk as D . Also note that $H^1 \cup_{i_2} D^2 \cong D$ is glued to $S^1 \times [0, 1]$ along a boundary piece $L \subset S^1 \times \{1\}$ that is also diffeomorphic to D^1 . We can have a diffeomorphism that shrinks the size of D relative to L so that the embedded D in M lies in a collar neighbourhood of the preimage of $S^1 \times \{1\}$. So the top boundary C in collar coordinates take the form of $\{(x, h(x)) | x \in S^1\}$ for some small and smooth h with $\text{supp}(h) \cong D^1$. Consider the a vector field on the collar neighbourhood with $v((x, h(x) + t)) = -g(t)h(x) \frac{\partial}{\partial t}$ with g being a one dimensional positive bump function with $g(0) = 1$. Flow everything in $S^1 \times [0, 1] \cup_L D^2$ with unit time, we get a diffeomorphism to $S^1 \times [0, 1]$ since by construction the image of C at time one will be $S^1 \times \{1\}$. Hence we get

$$\bigcup_{i=1}^n q^{-1}(v_i) \cong S^1 \times [0, 1] \cup_{S^1} D^2 \cong D^2$$

□

In general we have the handle cancellation theorem, that is attaching two handles with consecutive indices to the boundary of a manifold won't change its diffeotype.

3.2 Unique structure on $\mathbf{T}$

Corollary 2. *Suppose (M, A) is an compact, connected one dimensional manifold without boundary, then $(M, A) \cong (S^1, ST)$.*

Proof. Without boundary implies all open sets in A are homeomorphic to some open interval in $\mathbb{R}$. By compactness, take a finite cover of M , hence WLOG let $A = \{(U_i, f_i)\}_{i \in \{1, 2, \dots, n\}}$. We can assume n to be minimal in the sense of $U_i \cap U_j \cap U_k = \emptyset$ for $i \neq j, j \neq k, k \neq i$. We can reorder the indices so that U_i only intersects with U_{i-1} and U_{i+1} for $i \in \{2, \dots, n-1\}$, U_n intersects only with U_1, U_{n-1} , and U_1 intersects only with U_2, U_n . Hence glueing the intervals corresponding to the U_i in a cyclic order gives a diffeomorphism to (S^1, ST) . □

Lemma 7. *Let G be the Reeb graph of $\mathbf{T} \setminus \{\star\}$. For each edge $e \in G$, we have $q^{-1}(e)$ diffeomorphic to (S^1, ST) .*

Proof. To construct the Reeb graph we cut along regular value sets, and $\mathbf{T} \setminus \{\star\}$ has no boundary. Hence $f^{-1}(a_n)$ has no boundary points. Also by properness $f^{-1}(a_n)$ is compact, and $\mathbf{T}$ is two dimensional. Hence each edge in the graph corresponds to a connected compact one dimensional manifold without boundary, which is diffeomorphic to (S^1, ST) . □

Lemma 8. *Given any atlas A on $\mathbf{T} \setminus \{\star\}$, there exists some open $U \subset \mathbf{T} \setminus \{\star\}$ such that $U \cong D^2 \setminus \{0\}$ with $U \cup \{\star\}$ being a neighbourhood of $\{\star\}$ in $\mathbf{T}$.*

Proof. By the previous section, there exists some proper morse function f on $\mathbf{T} \setminus \{\star\}$, we have a sequence $a_n \uparrow \infty$ such that $f^{-1}(a_n)$ contains no critical points for all n . We claim that we have a subsequence b_n such that for all n , $f^{-1}(b_n)$ contains a circle that encircles the punctured point. Since if not, then there exists some $N > 0$ such that $f^{-1}([0, N])$ contains a disk neighbourhood of the punctured point, making it noncompact, contradicting to properness of f .

Consider its Reeb graph $G = q(\mathbf{T} \setminus \{\star\})$. Since the punctured torus retracts to $S^1 \vee S^1$. We have $\pi_1(\mathbf{T} \setminus \{\star\}) = \langle a, b \rangle$, so its fundamental group is finitely generated. Hence $\pi_1(G)$ induced by q is still finitely generated. Generators of $\pi_1(G)$ corresponds to finite cycles of G , which we have at most two of them. Hence we have a finite number of trees attached to the vertices on these cycles. Only one of the trees can be infinite and contains exactly one infinite tail homeomorphic to $[0, \infty)$. Since an infinite trees will contain at least one infinite tail with some finite trees attached, denote the union of the tail and the finite trees as K . Notice that the $q^{-1}(K)$ will contain all the circles in $f^{-1}(b_n)$ that encircles the punctured point, hence making it containing a disk neighbourhood of the point. Since q is surjective, the disk neighbourhood corresponds to K and the punctured torus with this neighbourhood removed is compact, hence corresponds to a finite graph in G . This shows two uniqueness statements. The K° will be the desired U , it is homeomorphic to an open punctured disk.

We can view $q^{-1}(K)$ differently, consider K . Notice that a finite tree always ends with a finite number of branching sticks. By the previous lemma the preimage of a branching stick is diffeomorphic to a cylinder. Hence, effectively we can eliminate all the vertices and edges corresponding to the stick in K . By iterations, the finite tree can eventually be removed from the infinite tail. We perform such operation to all finite trees attaching to the tail, making K homeomorphic to $[0, \infty)$. So $q^{-1}(K)$ is diffeomorphic to an infinite union of standard cylinders with boundaries glued smoothly. After taking the interior, we have $q^{-1}(K) \cong (S^1 \times [0, \infty))^\circ = S^1 \times (0, \infty)$. Notice that $(S^1 \times (0, \infty), \text{ST}) \cong (D^2 \setminus \{0\}, \text{ST})$. Hence the open punctured disk U with structure inherited from A is diffeomorphic to the standard punctured disk. $\square$

Lemma 9. *Every atlas on $\mathbf{T} \setminus \{\star\}$ can be extended to an atlas on $\mathbf{T}$*

Proof. We add the chart (U', i) to the atlas. Where $U' = U \cup \{\star\}$ and $i : U' \rightarrow D^2$ with $i(\star) = 0$ and $i|_U$ given by the diffeomorphism in the previous lemma. Notice that the transition maps are all identities. Hence we extend the differential structure to $\mathbf{T}$. $\square$

Theorem 10. *Smooth structure on $\mathbf{T}$ is unique*

Proof. Give $\mathbf{T}$ an atlas A . Since its Reeb graph $G = q((\mathbf{T}, A))$. We have $\pi_1 G$ being a quotient of $\pi_1 \mathbf{T} = \mathbb{Z}^2$. By compactness of $\mathbf{T}$, G is finite, the fundamental group of a finite connected graph is always free, the number of generators is the number of cycles within the graph. $\mathbb{Z}^2$ is not free, hence $\pi_1 G = 0$ or $\mathbb{Z}$. The first case is impossible since by elimination G can be reduced to G' where G' contains two vertices joined an edge. This gives a S^2 , since we are attaching two copies of D^2 to the top and the bottom boundary of $S^1 \times [0, 1]$, making it diffeomorphic to S^2 , contradiction since $S^2 \not\cong \mathbf{T}$. In the latter case, we have a circle shape graph with finitely many finite trees attached to it. Eliminate

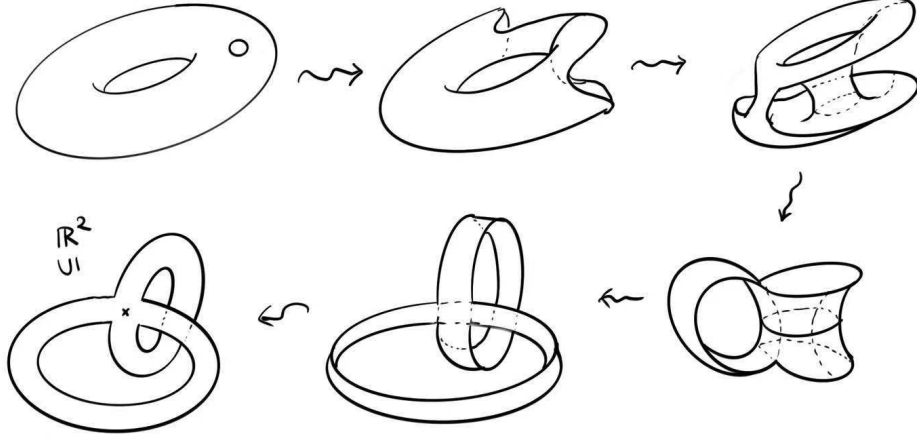


Figure 1: Immersion of the punctured torus into $\mathbb{R}^2$

all the trees we see cylinders glued smoothly in a cyclic manner. This gives $(\mathbf{T}, \text{ST})$ or the standard Klein bottle. But $(\mathbf{T}, A)$ is orientable hence

$$(\mathbf{T}, A) \cong (\mathbf{T}, \text{ST})$$

□

3.3 Smoothing theorems

Theorem 11. (*Vertex smoothing theorem*) For a smooth surface S with the atlas σ and $h_0 : (\mathbb{R}^2, ST) \rightarrow (S, \sigma)$ as a topological embedding, we can isotope h_0 to a smooth map h_1 near the origin and $h_1|_U = h_0|_U$ for some larger neighbourhood U of the origin.

Proof. We construct such isotopy h_t by defining it as $h_t = h_0 \circ G_t^{-1}$ for some isotopy $G_t : (\mathbb{R}^2, \chi) \rightarrow (\mathbb{R}^2, \text{ST})$ fixing the origin such that $G_0 = \text{id}$ and G_1 is smooth near the origin, and it is the identity outside some larger domain.

Since S is smooth, so the pull-back atlas χ_1 on $\mathbb{R}^2$ makes it naturally a smooth surface. Now consider the punctured torus $\mathbf{T} \setminus \{\star\}$, embed it in $\mathbb{R}^3$, expanding the punctured hole, we see an open disk D^2 attached by two strips, each homeomorphic to $((0, 1) \times (0, 1)) \cup (\{0, 1\} \times [0, 1])$, denote them by C_1, C_2 , embedding of $\mathbf{T} \setminus \{\star\}$ into $\mathbb{R}^2$ is impossible, but $D^2 \cup C_i$ s can individually embedded with the embeddings agree on D^2 , and via some suitable translations, assume the origin is covered, and hence by openness of $\mathbf{T} \setminus \{\star\}$, $B(\epsilon)$ is covered for some small $\epsilon > 0$.

Give them smooth structure by restrictions of χ_1 , and take the union which give the punctured torus an atlas χ_2 , (indeed taking the union is legit since they agree on the intersections). By the lemma the atlas extends to a new atlas χ_3 on $\mathbf{T}$. Again by the lemma we have a diffeomorphism

$$g : (\mathbf{T}, \chi_3) \rightarrow (\mathbf{T}, \text{ST})$$

g is also a homeomorphism, which induces an injective map

$$g_\star : \pi_1(\mathbf{T}) \cong \mathbb{Z}^2 \rightarrow \pi_1(\mathbf{T}) \cong \mathbb{Z}^2, \quad g_\star \in \text{GL}_2(\mathbb{Z})$$

however for any element $a \in \text{GL}_2(\mathbb{Z})$, it extends naturally to $A \in \text{GL}_2(\mathbb{R})$, and A clearly induces a diffeomorphism A on $(\mathbf{T}, \text{ST})$, but the differential structure on $\mathbf{T}$ is unique, hence it is also a diffeomorphism in $(\mathbf{T}, \chi_3)$. Hence we can WLOG assume $g_\star = I \in \text{GL}_2(\mathbb{Z})$. So we can lift g to a diffeomorphism $\hat{g} : (\mathbb{R}^2, \chi_4) \rightarrow (\mathbb{R}^2, \text{ST})$ (where χ_4 is the pull-back structure from $(\mathbf{T}, \chi_3)$) and having $\hat{g}$ fixing the lattice $\mathbb{Z}^2$, it is also periodic, that is $\hat{g}(x, y) = \hat{g}(x + n, y + m)$ for $(n, m) \in \mathbb{Z}^2$. Hence there exists some $C > 0$ such that $|\hat{g}(x, y) - (x, y)| < C$ for all $(x, y) \in \mathbb{R}^2$, since we certainly have this property over $[0, 1]^2$ by compactness, and by periodicity, we have the result on $\mathbb{R}^2$.

Now consider some diffeomorphism ϕ from $(\mathbb{R}^2, \text{ST})$ to the standard unit open disk $(B(1), \text{ST})$ such that it is the identity on some $B(\epsilon)$. In polar coordinates, we can define it as $\phi : (r, \theta) \mapsto (f(r), \theta)$ for some diffeomorphism $f : \mathbb{R}_{\geq 0} \rightarrow [0, 1]$ with $f(x) = x$ on $[0, \epsilon)$. χ_4 is pushed forward to a structure χ_5 on $B(1)$, and they match exactly near the origin, since f is the identity there. Hence $\hat{G} = \phi \circ \hat{g} \circ \phi^{-1} : (B(1), \chi_5) \rightarrow (B(1), \text{ST})$ a diffeomorphism. We can extend it continuously to a homeomorphism $G : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that is the identity on $\mathbb{R} \setminus B(1)$. It is continuous since for $x, y \in \mathbb{R}^2$, we have $|\phi(x) - \phi(y)| \rightarrow 0$ when $|x| \rightarrow \infty$ for all y such that $|x - y| < C$. After the extension, we have G^{-1} a diffeomorphism on $B(1)$. Notice that it doesn't make sense to say whether G^{-1} is a diffeomorphism at $x \in \mathbb{R}^2 \setminus B(1)$ since we don't know whether χ_5 can be extended to $\mathbb{R}^2$ or not, but it is acceptable since we only care about smoothing near the origin, and a homeomorphism else where will be sufficient. So far we've constructed the following maps

$$\begin{array}{ccccccccc} \mathbb{R}^2 & \xrightarrow{i_3} & (B(1), \chi_5) & \xrightarrow{\phi^{-1}} & (\mathbb{R}^2, \chi_4) & \xrightarrow{q} & (\mathbf{T}, \chi_3) & \xrightarrow{i_2} & (\mathbf{T} \setminus \{\star\}, \chi_2) & \xrightarrow{i_1} & (\mathbb{R}^2, \chi_1) \\ \downarrow G & & \downarrow \hat{G} & & \downarrow \hat{g} & & \downarrow g & & & & \downarrow h_0 \\ (\mathbb{R}^2, \text{ST}) & \xrightarrow{i_3} & (B(1), \text{ST}) & \xleftarrow[\phi]{} & (\mathbb{R}^2, \text{ST}) & \xrightarrow{q} & (\mathbf{T}, \text{ST}) & & & & (S, \sigma) \end{array}$$

Where i_3 denotes the restriction. All the maps are continuous, hence pick some $\delta > 0$ small enough such that

$$(i_1 \circ i_2 \circ q \circ \phi^{-1} \circ i_3 \circ G^{-1})(B(\delta)) \subset B(\epsilon) \subset i_1(\mathbf{T} \setminus \{\star\})$$

by viewing G^{-1} as a map from $(\mathbb{R}^2, \text{ST})$ to $(\mathbb{R}^2, \chi_1)$, it will be a diffeomorphism on $B(\delta)$ (or near the origin). Hence $h \circ G^{-1}$ is smooth near the origin. We can define G_t by the following

$$G_t|_{tB(1)}(x) = tG(x/t) \subset B(1), \quad G_t|_{\mathbb{R}^2 \setminus tB(1)} = \text{id for } t \in (0, 1], \quad G_0 = \text{id}$$

we can check that $G_1 = G$, hence by letting $h_t = h_0 \circ G_t^{-1}$ is yields the desired isotopy. $\square$

Lemma 10. $(D^1 \times \mathbb{R}, A) \cong (D^1 \times \mathbb{R}, ST)$ for an atlas A standard near $\partial D^1 \times \mathbb{R}$. The diffeomorphism can be chosen to the identity near $\partial D^1 \times \mathbb{R}$ (by near, we mean on an open neighbourhood of it).

Proof. Structure being standard near the boundary implies the projection map $p : D^1 \times \mathbb{R} \rightarrow \mathbb{R}$ is morse and proper near the boundary $\partial D^1 \times \mathbb{R}$. Perturbations in the interior can make it into a proper morse function on $D^1 \times \mathbb{R}$. We can glue two copies of $D^1 \times \mathbb{R}$ to get a $S^1 \times \mathbb{R}$ and a proper morse function on it. Notice that $S^1 \times \mathbb{R}$ has no boundary, hence we have edges in its Reeb graph also corresponds to S^1 . Also the graph consists of a infinite line homeomorphic to $\mathbb{R}$ with finite trees attached to it. We can perform the branching stick elimination process to it and only left with the infinite line. This corresponds to $(S^1 \times \mathbb{R}, ST)$. Notice that the diffeomorphisms on $S^1 \times \mathbb{R}$ induced by branching stick elimination is the identity near $\partial D^1 \times \mathbb{R} \subset S^1 \times \mathbb{R}$ since all critical points away from it. (An alternative way to see this is to consider the flow generated by f , which is $\frac{\partial}{\partial t}$ near the boundary. Hence starting from any $(x, 0)$ for x near ∂D^1 , we will reach (x, t) at time t). Restricting to $D^1 \times \mathbb{R} \subset S^1 \times \mathbb{R}$, we get a diffeomorphism from $(D^1 \times \mathbb{R}, A)$ to $(D^1 \times \mathbb{R}, ST)$ that is the identity near $\partial D^1 \times \mathbb{R}$. $\square$

Theorem 12. (edge smoothing theorem) For a smooth surface S with the atlas σ and $h_0 : (D^1 \times \mathbb{R}, ST) \rightarrow (S, \sigma)$ as a topological embedding with h_0 smooth near $\partial D^1 \times \mathbb{R}$, we can isotope h_0 to a smooth map h_1 near $D^1 \times \{0\}$ and $h_1|_U = h_0|_U$ for some larger neighbourhood U of $D^1 \times \{0\}$ (here we assume $D^1 = [-1, 1]$).

Proof. Similarly we will consider some $G_t : (D^1 \times \mathbb{R}, ST) \rightarrow (D^1 \times \mathbb{R}, \chi_1)$ with χ_1 being the pull-back atlas from h_0 . h_0 being smooth in the standard sense near the boundary implies χ_1 agrees with the standard structure near $\partial D^1 \times \mathbb{R}$, by the previous lemma, we have some diffeomorphism $g : (D^1 \times \mathbb{R}, \chi_1) \rightarrow (D^1 \times \mathbb{R}, ST)$ that is the identity near the boundary. Consider some diffeomorphism ϕ from $(D^1 \times \mathbb{R}, ST) \rightarrow (C, ST)$ for $C = (D^1 \times D^1) \setminus (\{0\} \times \partial D^1)$ with $\phi = \text{id}$ on $D^1 \times [-\delta, \delta]$. We can pull back the atlas χ_1 to χ_2 on C , making the map $\hat{G} = \phi^{-1} \circ g \circ \phi$ a diffeomorphism from (C, χ_2) to (C, ST) . Since the boundary of $D^1 \times \mathbb{R}$ is mapped to $\partial(D^1 \times D^1)$ with only two points being removed. We can extend $\hat{G}$ to a homeomorphism G from $D^1 \times \mathbb{R}$ to itself by the identity outside $D^1 \times D^1$ and on $\{0\} \times \partial D^1$. We have the following diagram

$$\begin{array}{ccccccc} D^1 \times \mathbb{R} & \xrightarrow{i} & (C, \chi_2) & \xrightarrow{\phi^{-1}} & (D^1 \times \mathbb{R}, \chi_1) & \xrightarrow{h_0} & (S, \sigma) \\ \downarrow G & & \downarrow \hat{G} & & \downarrow g & & \\ (D^1 \times \mathbb{R}, ST) & \xrightarrow{i} & (C, ST) & \xleftarrow[\phi]{} & (D^1 \times \mathbb{R}, ST) & & \end{array}$$

where i denotes the restriction. View G as a map from $(D^1 \times \mathbb{R}, ST)$ to $(D^1 \times \mathbb{R}, \chi_1)$. We naturally will have it being a diffeomorphism on $D^1 \times (-\delta, \delta)$. Hence $h_0 \circ G^{-1}$ is a diffeomorphism near $D^1 \times \{0\}$. To get the isotopy, we define

$$G_t|_{tC}(x) = tG(x/t) \subset C, \quad G_t|_{D^1 \times \mathbb{R} \setminus C} = \text{id for } t \in (0, 1], \quad G_0 = \text{id}$$

Hence $h_t = h_0 \circ G_t^{-1}$ yields the desired isotopy. $\square$

Lemma 11. $(D^2, A) \cong (D^2, ST)$ for an atlas A standard near $\partial D^1 \times \mathbb{R}$. The diffeomorphism can be chosen to be the identity near S^1 .

Proof. Consider the function $f : (r, \theta) \rightarrow r$ in polar coordinates. The standard structure near S^1 makes it a proper morse function near the boundary. Similary, perturbation in $B(1)$ can make it into a proper morse function on D^2 . We can glue two copies of D^2 smoothly on S^1 to get a proper morse function. Its Reeb graph is a finite tree, notice that $f^{-1}(1) = S^1$ and all other level sets are inside $B(1)$, $B(1)$ has no boundary, making them disjoint unions of circles. So we can perform branching stick elimination, and eventually get a single edge with only a single vertex, which corresponds to the glueing of a standard disk D^2 to S^1 , this is just (D^2, ST) . $\square$

Theorem 13. (face smoothing theorem) For a smooth surface S with the atlas σ and $h_0 : (D^2, ST) \rightarrow (S, \sigma)$ as a topological embedding with h_0 smooth near S^1 , we can isotope h_0 to a smooth map h_1 near 0 and $h_1|_U = h_0|_U$ for some larger neighbourhood U of 0.

Proof. By smoothness of h_0 , the pull-back structure χ_1 on D^2 is standard near S^1 . By the previous lemma we have some diffeomorphism $G : (D^2, ST) \rightarrow (D^2, \chi_1)$ that is the identity near S^1 . Define G_t as

$$G_t|_{tD^2} = tG(x/t), \quad G_t|_{D^2 \setminus tD^2} = \text{id}, \quad G_0 = \text{id}$$

and let $h_t = h_0 \circ G_t^{-1}$. $\square$

Finally we are in the position to show the required theorem.

Theorem 14. homeomorphism between two smooth triangulable surfaces can be isotoped into a diffeomorphism.

Proof. For a homeomorphism $f : (S, \sigma) \rightarrow (S', \sigma')$. Triangulate S and push the triangulation to S' . Let p be a vertex in S and p' be its image. Consider two local parametrisations $\phi : U \rightarrow S$, $\phi' = f \circ \phi : U \rightarrow S'$ with $p \in \phi(U)$. We have ϕ' being a topological embedding into S' , hence by the vertex smoothing theorem, we can isotope ϕ' to a diffeomorphism φ' . Consider the map $F = \varphi' \circ \phi^{-1} : \phi(U) \subset S \rightarrow S'$. Notice that $\phi'^{-1} \circ F \circ \phi = \phi'^{-1} \circ \varphi' \circ \phi^{-1} \circ \phi = \phi'^{-1} \circ \varphi$. Since $\varphi' : U \rightarrow S'$ is a diffeomorphism, by definition we have $\varphi' \circ \phi'^{-1}$ a diffeomorphism, hence $F|_{\phi(U)} : \phi(U) \subset S \rightarrow S'$ is a diffeomorphism. We can perform the above operation to every vertex, hence we can isotope f into f_1 that is a local diffeomorphism near every vertex.

By definition of being triangulable, we can enlarge the domain of ϕ to contain $[0, 1]$, with $\phi(0), \phi(1)$ being two vertices. We define $\phi' = f_1 \circ \phi$. By construction, ϕ' is a topological embedding, but smooth near 0 and 1. By applying the edge smoothing theorem, we can isotope ϕ' to φ' , and the rest steps are the same as above. Hence we can isotope f_1 to f_2 that is a diffeomorphism near every edge.

For the faces, further enlarge the domain to contain some triangle T , and $\phi' = f_2 \circ \phi$ satisfies all the conditions in the face smoothing theorem. By following similar procedures, we see that f_2 can be isotoped into a global diffeomorphism f_3 .



4 References

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